

Evaluation metrics for Regression

Regression evaluation

- The most commonly used **evaluation measures for regression models** are:
 1. Mean Absolute Error (MAE)
 2. Mean Squared Error (MSE)
 3. Root Mean Squared Error (RMSE)
 4. Coefficient of Determination (R^2 Score)
 5. Residual Sum of Squares (RSS or SSE)

1. Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- y_i = Actual (observed) value of the dependent variable.
 - $\hat{y}_i$ = Predicted value of the dependent variable obtained from the regression model.
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- Lower value indicates better performance.
 - Less sensitive to outliers than MSE.

2. Mean Squared Error (MSE)

Measures the average squared prediction error.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Penalizes large errors more heavily.
- Widely used as the loss function in regression.

3. Root Mean Squared Error (RMSE)

Square root of MSE.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- Expressed in the same units as the target variable.
- Easier to interpret than MSE.

4. Coefficient of Determination (R^2 Score)

Measures the proportion of variance in the dependent variable explained by the model.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

where:

- $\bar{y}$ = mean of the actual values
- $R^2 = 1$: Perfect fit.
- $R^2 = 0$: Model performs no better than predicting the mean.
- $R^2 < 0$: Model performs worse than predicting the mean.

5. Residual Sum of Squares (RSS or SSE)

Measures the total squared prediction error.

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Lower RSS indicates a better fit.